

System Capacity & Care Load Analytics for Unaccompanied Children

Business Analytics Internship Project

Prepared By

Rupesh Sunilrao Malkapure

Role: Business Analyst Intern

Project: System Capacity & Care Load Analytics for Unaccompanied Children

Internship Project Report

Table of Content

Sr.no		Page.no
	EXECUTIVE SUMMARY	1
	ABSTRACT	1
1.	INTRODUCTION	2-3
2.	PROBLEM STATEMENT AND RESEARCH OBJECTIVES	3-4
3.	DATASET DESCRIPTION	4-5
4.	RESEARCH METHODOLOGY	5-6
5.	DATA PREPARATION AND FEATURE ENGINEERING	7-9
6.	RESULTS AND ANALYSIS	9-13
7.	KEY FINDINGS	13-14
8.	RECOMMENDATION	14
9.	LIMITATIONS AND FUTURE SCOPE	14-15
10.	CONCLUSION	15-16

EXECUTIVE SUMMARY

This study, titled “System Capacity & Care Load Analytics for Unaccompanied Children,” examines operational load patterns associated with children moving through U.S. Customs and Border Protection (CBP) custody and Health and Human Services (HHS) care. The primary objective of the study is to analyse system capacity, care load, daily intake patterns, and periods of sustained operational strain using time-series operational data covering the period from 12 January 2023 to 21 December 2025.

The analysis was conducted using Python and Pandas for data preparation, transformation, and feature engineering, while Tableau was used to develop an interactive analytical dashboard. Key analytical measures included System Load, Net Daily Intake, High Load, Sustained High Load, Prolonged Strain, 7-day and 14-day moving averages, and Backlog Accumulation Rate. High Load was defined as a System Load greater than the overall mean, while duration-based measures were used to distinguish temporary increases from sustained periods of operational pressure.

The analysis found that the average System Load was approximately 6,233 children. The maximum System Load reached 11,762 children on 20 December 2023, while the minimum was 2,002 children on 24 August 2025. The average number of children in CBP custody was 171.5, compared with 6,061 in HHS care, indicating that HHS care represented the substantially larger component of the overall measured system load.

The average Net Daily Intake was -44.74 children, with a maximum value of +206 on 12 February 2024 and a minimum value of -465 on 11 January 2024. The analysis further identified 429 high-load observations and 411 prolonged-strain observations, showing that elevated system load frequently occurred within sustained periods rather than only as isolated fluctuations. The Tableau analysis also produced an aggregated Backlog Accumulation Rate of 33.06% under the applied table-calculation settings; this value represents the aggregated calculated rate in the dashboard view and should not be interpreted as total backlog growth over the entire study period.

Overall, the project demonstrates how operational data can be transformed into meaningful indicators for monitoring changes in system load and identifying periods of sustained pressure. The interactive Tableau dashboard integrates the major indicators into a single analytical view, allowing users to examine changing operational patterns over time. The project demonstrates the practical application of Python, Pandas, time-series analysis, feature engineering, and Tableau dashboard development to an operational analytics problem.

ABSTRACT

The movement of children through U.S. Customs and Border Protection (CBP) custody and Health and Human Services (HHS) care creates varying levels of operational pressure that require continuous monitoring of custody, transfers, intake, and discharge patterns. This study applies data analytics and visualization techniques to examine system load, backlog dynamics, and periods of sustained strain using daily operational data covering the period from 12 January 2023 to 21 December 2025.

Python and Pandas were used for data preparation, transformation, and feature engineering, while Tableau was employed to develop an interactive analytical dashboard. The analysis incorporates key operational measures related to CBP custody and HHS care and derives additional indicators including total system load, net daily intake, high-load periods, sustained high-load duration, 7-day and 14-day moving averages, and prolonged system strain.

The analysis recorded an average system load of 2,502 children, while the maximum observed system load reached 11,762 children on 20 December 2023. The average net daily intake was -44.74 children per day, indicating that, on average, the net intake measure showed a negative daily balance over the observed records. Furthermore, 429 high-load days and 411 prolonged-strain days were identified, demonstrating that elevated operational pressure persisted across a substantial number of observations in the study period.

The study provides a structured analytical framework for identifying short-term fluctuations, longer-term load patterns, and periods of sustained operational strain within the child custody system. The interactive dashboard further enables dynamic examination of operational metrics over time, supporting systematic monitoring and evidence-based interpretation of changing custody conditions.

Keywords: Child Custody Analytics, CBP, HHS, System Load, Backlog Analytics, Net Daily Intake, High Load, Prolonged Strain, Moving Average, Python, Tableau.

1. INTRODUCTION

The management of children within the United States immigration custody system involves coordination between multiple federal agencies, particularly U.S. Customs and Border Protection (CBP) and the Department of Health and Human Services (HHS). Children entering the system may initially remain in CBP custody before being transferred to HHS care, while subsequent discharges from HHS continuously change the number of children present within the overall care system. These movements create considerable variation in system load over time.

Daily custody figures alone may not provide a complete understanding of operational pressure. A sudden increase in the number of children in care may represent a temporary fluctuation, whereas consistently elevated levels over several consecutive days can indicate sustained system strain. Similarly, differences between incoming and outgoing movements can contribute to backlog accumulation and affect the overall capacity of the system. Therefore, analysing both the magnitude and duration of changes is important for understanding operational patterns.

Time-series analytics provides an effective approach for examining these dynamics. Measures such as total system load and net daily intake help quantify overall pressure and daily changes, while moving averages can reduce short-term fluctuations and reveal underlying trends. Duration-based indicators can further distinguish isolated high-load observations from sustained periods of elevated pressure.

The present study analyses daily child custody data covering the period from **12 January 2023 to 21 December 2025**. The study integrates CBP and HHS operational measures to examine changes in total system load, net intake, backlog patterns, high-load periods, and prolonged strain. Python and Pandas are used for data preparation and analytical feature development, while Tableau is used to construct an interactive dashboard for visual exploration of the resulting patterns.

The study aims to transform daily operational records into interpretable indicators that provide a clearer understanding of how custody pressure changes over time. By combining trend analysis, moving averages, load classification, and duration-based measures, the research develops a structured framework for identifying both short-term fluctuations and sustained periods of operational pressure.

2. PROBLEM STATEMENT AND RESEARCH OBJECTIVES

2.1 Problem Statement

The daily movement of children through CBP custody and HHS care can result in substantial fluctuations in the overall number of children within the custody system. Although daily operational data provide information on apprehensions, custody levels, transfers, and discharges, individual figures alone may not clearly indicate whether the system is experiencing temporary variation or sustained operational pressure.

A rise in custody levels can occur when incoming movements exceed the system's ability to transfer or discharge children, while continued elevated levels may contribute to persistent system strain. Therefore, analysing only individual daily values may overlook broader patterns such as backlog accumulation, consecutive high-load periods, and longer-term changes in system pressure.

The analytical problem addressed in this study is therefore to transform daily CBP and HHS operational records into meaningful indicators that can identify changes in system load, intake patterns, backlog behaviour, and sustained periods of operational strain. The study also seeks to provide an interactive visual framework through which these patterns can be examined across different time periods.

2.2 Research Objectives

The study has the following objectives:

1. To analyse the trend and variation in overall child custody system load across CBP custody and HHS care during the study period.
2. To compare CBP custody and HHS care levels and examine their contribution to overall system pressure.
3. To evaluate net daily intake and backlog patterns to understand changes in the balance between incoming and outgoing movements.

4. To identify high-load periods, sustained high-load conditions, and prolonged system strain using duration-based analytical indicators.
5. To examine short-term and underlying system-load trends using 7-day and 14-day moving averages.
6. To develop an interactive Tableau dashboard for monitoring key system-load indicators and analysing changes across different time periods.

3. DATASET DESCRIPTION

3.1 Dataset Overview

The study is based on daily operational data related to children moving through U.S. Customs and Border Protection (CBP) custody and Health and Human Services (HHS) care. The original dataset consists of **1,170 observations and six variables**, covering operational records within the study period from **12 January 2023 to 21 December 2025**. The dataset contains **720 unique dates**, providing a multi-year basis for analysing changes in custody levels and system pressure.

The dataset captures different stages of the child custody and care process, including apprehension and placement into CBP custody, the number of children remaining in CBP custody, transfers out of CBP custody, children present in HHS care, and children discharged from HHS care.

3.2 Original Variables

The six original variables used in the study are:

Variable	Description
Date	Date associated with the daily operational record
Children apprehended and placed in CBP custody	Number of children apprehended and initially placed under CBP custody
Children in CBP custody	Number of children remaining in CBP custody
Children transferred out of CBP custody	Number of children transferred out of CBP custody
Children in HHS Care	Number of children under HHS care
Children discharged from HHS Care	Number of children discharged from HHS care

3.3 Analytical Scope of the Dataset

The original variables provide information about different stages of movement through the custody and care system. However, these raw measures alone are insufficient for identifying sustained pressure, backlog patterns, and longer-term trends. Therefore, additional analytical variables were subsequently developed, including System Load, Net Daily Intake, High Load,

High Load Duration, Sustained High Load, 7-Day Moving Average, 14-Day Moving Average, and Prolonged Strain.

These derived indicators enable the study to move beyond descriptive daily counts and evaluate the magnitude, direction, and persistence of operational pressure over time.

4. RESEARCH METHODOLOGY

4.1 Research Design

The study adopts a **descriptive and exploratory data analytics approach** to examine temporal patterns in child custody system load, daily intake, backlog behaviour, and sustained operational strain. The analysis is based on historical daily operational records and focuses on identifying both short-term fluctuations and persistent patterns over the study period.

A time-series-oriented analytical framework was used because the observations are associated with dates and the primary objective is to understand how system conditions change over time.

4.2 Data Processing and Preparation

The dataset was processed using **Python and the Pandas library**. Initial data preparation involved examining the structure of the dataset, identifying missing values, verifying data types, and converting relevant variables into appropriate numerical formats.

Numeric fields containing formatting characters such as commas were cleaned before conversion to numeric data types. Date information was also structured appropriately to support chronological analysis. Missing observations were retained where necessary to preserve the continuity of the calendar-based time series rather than removing dates solely because operational values were unavailable.

4.3 Feature Engineering

Several derived variables were developed from the original operational measures to provide a more meaningful assessment of system pressure.

System Load was calculated by combining the number of children in CBP custody and the number of children in HHS care:

$$\text{System Load} = \text{Children in CBP Custody} + \text{Children in HHS Care}$$

Net Daily Intake was used to examine the daily balance between incoming and outgoing movements within the operational system.

Additional indicators were developed to classify **High Load** conditions and identify consecutive periods during which elevated system load persisted. Consecutive observations were grouped to calculate **High Load Duration**, from which **Sustained High Load** conditions were identified.

4.4 Moving-Average Analysis

To distinguish short-term fluctuations from broader trends, **7-day and 14-day moving averages** of system load were calculated.

For a moving window of n observations:

$$MA_n(t) = \frac{1}{n} \sum_{i=0}^{n-1} SystemLoad_{t-i}$$

The 7-day moving average provides greater sensitivity to recent changes, while the 14-day moving average provides a smoother representation of the underlying system-load trend.

4.5 Strain and Duration Analysis

Duration-based analysis was used to distinguish isolated high-load observations from persistent operational pressure. Consecutive high-load observations were grouped and their duration was measured. This approach was further extended to identify periods of **prolonged strain**, representing sustained pressure rather than a temporary increase in daily load.

This distinction is important because a single high-load observation and an extended period of elevated load may have different operational implications.

4.6 Data Visualization and Dashboard Development

Following data preparation and feature engineering, **Tableau** was used to develop an interactive analytical dashboard. The dashboard incorporated system-load trends, CBP and HHS load comparisons, net daily intake and backlog patterns, high-load and prolonged-strain indicators, and key performance indicators (KPIs).

Interactive functionality was incorporated to allow users to examine operational patterns across different dates, metrics, and time periods. This combination of Python-based analytical processing and Tableau-based visualization provided a structured workflow from raw operational data to interpretable analytical insights.

5. DATA PREPARATION AND FEATURE ENGINEERING

Data preparation and feature engineering were performed to transform the original operational records into analytical indicators capable of representing system pressure, short-term changes, and sustained strain. Python and Pandas were primarily used for this stage of the analysis.

5.1 Data Cleaning and Type Conversion

The dataset was initially examined for data types, missing observations, and formatting inconsistencies. Some numerical variables contained comma-separated values and were therefore stored as text rather than numeric data. These formatting characters were removed and the relevant variables were converted into numeric form before performing arithmetic calculations.

Missing operational values were handled carefully. Dates were retained where appropriate to maintain the chronological structure of the dataset rather than deleting complete calendar observations solely because individual operational measures were unavailable.

5.2 System Load

A combined System Load indicator was developed to represent the total number of children present across the two major components of the analysed care system:

$$System\ Load_t = CBP\ Custody_t + HHS\ Care_t$$

This variable provides a consolidated measure of overall system pressure and serves as one of the principal indicators used throughout the analysis.

5.3 Net Daily Intake

Net Daily Intake was developed to measure the balance between incoming and outgoing movements on a daily basis. This indicator helps determine whether the operational system is experiencing net accumulation or net reduction on a given observation.

Positive values indicate that incoming movements exceeded outgoing movements, whereas negative values indicate that outgoing movements exceeded incoming movements.

5.4 High-Load Classification

A High Load indicator was created to classify observations according to whether system load exceeded the defined analytical threshold. The resulting Boolean classification identifies each observation as either a high-load or non-high-load condition.

This classification provides the foundation for analysing not only how frequently high-load conditions occurred but also whether they persisted across consecutive observations.

5.5 Consecutive High-Load Groups and Duration

To identify continuous periods of similar load conditions, changes between the current high-load status and the previous observation were detected. Each change initiated a new sequential group.

The grouping logic can be represented as:

$$Group_t = CumulativeSum(HighLoad_t \neq HighLoad_{t-1})$$

The duration of each high-load group was subsequently calculated. This made it possible to distinguish an isolated high-load observation from a continuous episode of elevated system pressure.

5.6 Sustained High Load

A Sustained High Load indicator was developed by combining the high-load classification with duration information. High-load conditions persisting for three or more consecutive observations were classified as sustained high-load periods.

This variable provides a stronger indication of persistent operational pressure than the high-load indicator alone.

5.7 Moving Averages

Two moving-average indicators were developed using System Load:

$$\begin{aligned} 7\text{-Day Average} &= MA_7(\text{System Load}) \\ 14\text{-Day Average} &= MA_{14}(\text{System Load}) \end{aligned}$$

The 7-day average captures relatively recent movements in system load, whereas the 14-day average smooths a larger portion of short-term variation and provides a clearer view of the underlying trend.

5.8 Prolonged Strain

A Prolonged Strain indicator was developed to identify periods in which system pressure remained elevated for an extended sequence of observations. Strain groups and their corresponding durations were used to separate temporary pressure from persistent operational strain.

This measure complements the high-load indicators by emphasizing the persistence of elevated system conditions.

5.9 Backlog Accumulation Rate

To assess the rate at which overall system load changed relative to the previous observation, a Backlog Accumulation Rate was used:

$$Backlog\ Accumulation\ Rate_t = \frac{SystemLoad_t - SystemLoad_{t-1}}{SystemLoad_{t-1}} \times 100$$

A positive rate indicates an increase relative to the preceding observation, while a negative rate indicates a reduction. This measure was incorporated into the Tableau analysis to provide a percentage-based indication of short-term changes in system load.

6. RESULTS AND ANALYSIS

6.1 Overall System Load Analysis

The system load analysis was conducted to evaluate the overall number of children simultaneously present across CBP custody and HHS care during the study period. System load represents the combined operational burden across these two components and therefore provides an overall measure of pressure within the analysed custody and care system.

The results indicate that the average system load was 2,502 children during the study period. However, considerable variation was observed across individual dates. The maximum system load reached 11,762 children on 20 December 2023, representing the highest recorded level in the analysed data. In comparison, the minimum system load was 2,002 children on 24 August 2025.

System Load Indicator	Result
Average System Load	2,502 children
Maximum System Load	11,762 children
Date of Maximum Load	20 December 2023
Minimum System Load	2,002 children
Date of Minimum Load	24 August 2025

The substantial difference between the maximum and minimum values demonstrates that system pressure was not constant throughout the study period. The peak recorded in December 2023 was substantially above the overall average, indicating a period of unusually elevated system load. Conversely, the minimum observed in August 2025 was below the overall average.

The 7-day and 14-day moving averages incorporated into the analysis further support the examination of these changes by smoothing short-term daily fluctuations. These measures allow isolated spikes to be distinguished from more persistent changes in the underlying system-load trend.

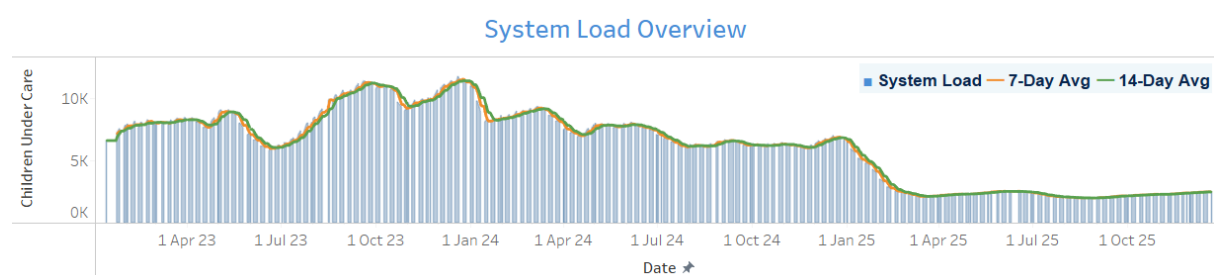


Figure 1: Trend in System Load with 7-Day and 14-Day Moving Averages, 2023–2025

The system-load trend and its moving averages therefore provide the foundation for the subsequent analysis of custody distribution, net intake, high-load conditions, and prolonged system strain.

6.2 CBP vs HHS Load Comparison

The comparison between CBP custody and HHS care reveals a substantial difference in their average daily care levels during the study period. The average number of children in CBP custody was approximately 171.5, whereas the average number of children in HHS care was 6,061.

Component	CBP Custody	HHS Care	Overall System Load
Average Children	171.5	6,061	6,233

The results indicate that HHS care accounted for the substantially larger component of the overall system load during the analysed period. The combined average system load of approximately 6,233 children is consistent with the contribution of the two components.

This difference highlights the importance of analysing CBP custody and HHS care separately rather than relying exclusively on the combined system-load indicator. While overall system load measures total operational pressure, the component-level comparison helps identify where children were predominantly present within the analysed system.

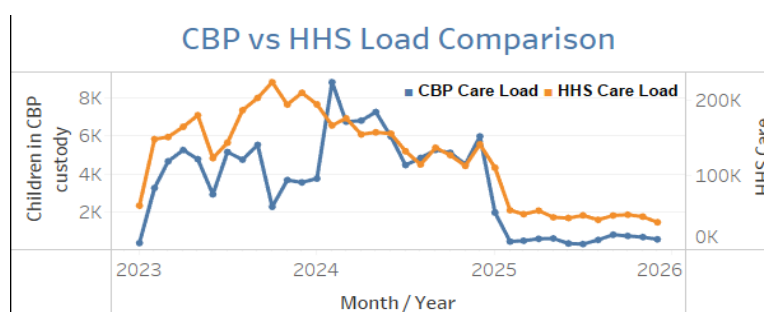


Figure 2: Comparison of Average Children in CBP Custody and HHS Care

6.3 Net Daily Intake and Backlog Analysis

Net Daily Intake was analysed to understand the daily balance of children entering and leaving the operational system. During the study period, the average Net Daily Intake was -44.74 children per day. The highest observed value was $+206$ on 12 February 2024, while the lowest value was -465 on 11 January 2024.

The negative overall average indicates that, across the analysed observations, the daily net balance was more frequently weighted toward reductions than additions. However, the substantial variation between the maximum and minimum values demonstrates that the system experienced considerable short-term fluctuations in daily intake and outflow.

The maximum positive Net Daily Intake of 206 represents a day when additions exceeded corresponding outflows according to the constructed measure. In contrast, the minimum value of -465 represents a substantially larger net reduction. These fluctuations are important for operational analysis because sharp positive changes can contribute to increasing system load, while negative values can contribute to reducing existing load.

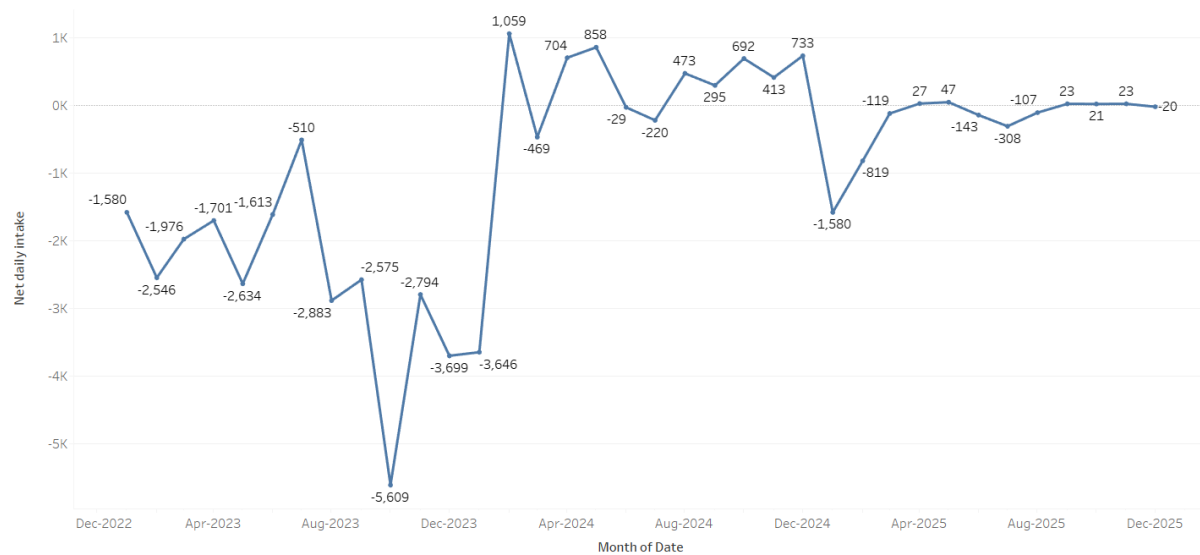


Figure 3: Trend in Net Daily Intake

6.4 High Load and Prolonged Strain Analysis

To identify periods of elevated operational pressure, each observation was classified as either a high-load or normal-load observation. A high-load condition was defined as a System Load greater than the mean System Load of the dataset. The average System Load during the study period was approximately 6,233 children, which therefore served as the analytical reference for identifying above-average load conditions.

The analysis identified 429 high-load observations, indicating that system load exceeded its overall mean on a substantial number of observations during the study period. However, an isolated high-load observation does not necessarily represent sustained operational pressure. Therefore, consecutive high-load observations were further analysed using duration-based grouping.

A period was classified as Sustained High Load when the high-load condition continued for at least three consecutive observations. The duration-based analysis was further used to identify prolonged periods of system strain. Overall, 411 observations were classified as Prolonged Strain, demonstrating that much of the identified elevated load occurred within sustained periods rather than only as isolated spikes.

These results demonstrate the value of combining a simple load threshold with duration-based indicators. While the High Load measure identifies observations with above-average system

load, the Prolonged Strain measure helps distinguish persistent operational pressure from temporary fluctuations.

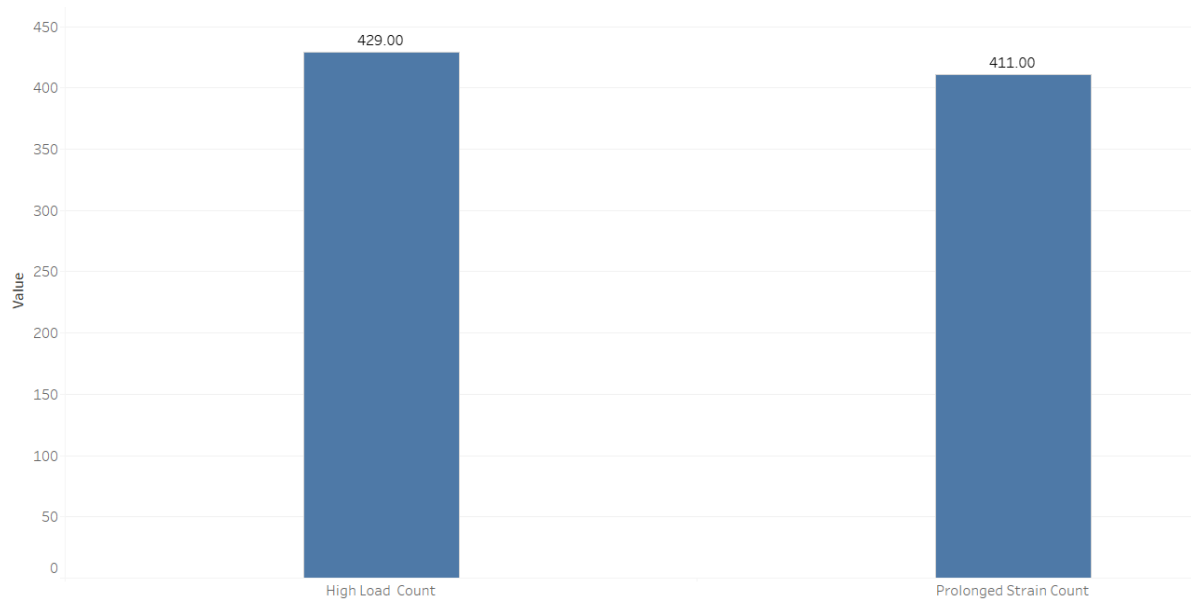


Figure 4: High Load and Prolonged Strain Patterns

6.5 Backlog Accumulation Rate Analysis

The Backlog Accumulation Rate was used to examine changes in System Load relative to the preceding observation. The indicator was calculated as:

$$\text{Backlog Accumulation Rate} = \frac{\text{Current System Load} - \text{Previous System Load}}{\text{Previous System Load}} \times 100$$

The Tableau analysis produced an aggregated Backlog Accumulation Rate of 33.06%. The positive value indicates an overall positive accumulation pattern under the aggregation applied in the dashboard calculation.

However, this value should be interpreted carefully because the Backlog Accumulation Rate is based on period-to-period percentage changes and Tableau displays it as an aggregated table calculation. Therefore, the 33.06% value should not be interpreted as meaning that the total backlog increased by exactly 33.06% over the entire study period. Instead, it represents the aggregated result of the calculated accumulation-rate measure under the current Tableau view and aggregation settings.

The indicator complements Net Daily Intake and System Load analysis by showing the relative direction and magnitude of changes in operational load between consecutive observations.

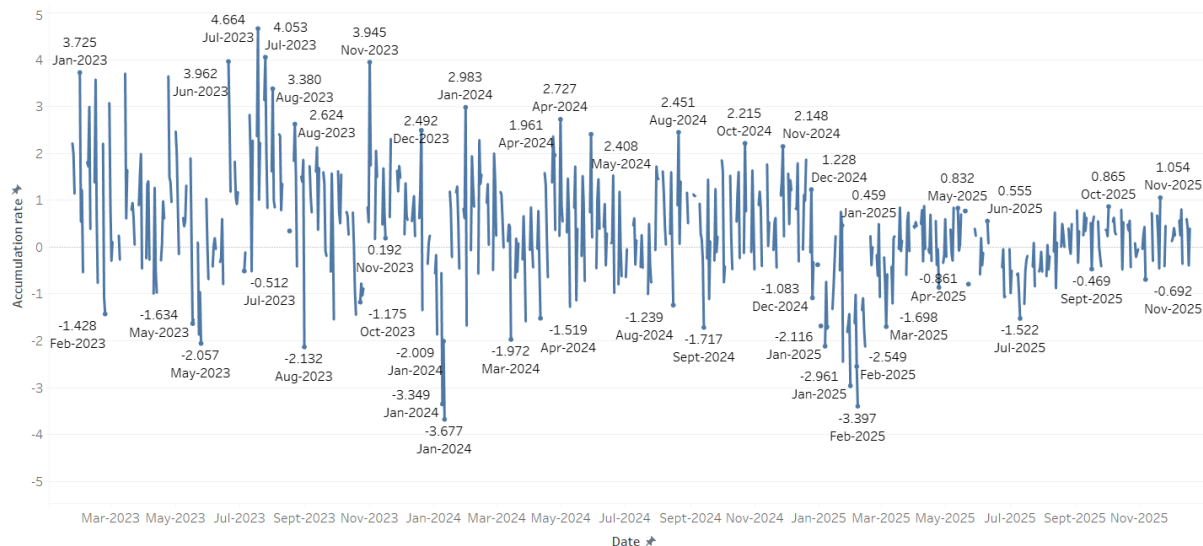


Figure 5: Backlog Accumulation Rate Trend

7. KEY FINDINGS

The analysis of U.S. child custody operational data produced several important findings regarding system load, daily intake patterns, and sustained operational strain.

1. The average System Load was approximately 6,233 children, while the maximum System Load reached 11,762 children on 20 December 2023. The minimum recorded System Load was 2,002 children on 24 August 2025.
2. A clear difference was observed between the two major components of the system. The average number of children in CBP custody was 171.5, compared with 6,061 children in HHS care, showing that HHS care represented the substantially larger component of the measured System Load.
3. The average Net Daily Intake was -44.74 children. The maximum Net Daily Intake was $+206$ on 12 February 2024, whereas the minimum was -465 on 11 January 2024, demonstrating considerable short-term variation in the daily balance.
4. Using the overall mean System Load as the threshold, 429 observations were classified as High Load. Duration-based analysis further identified 411 observations as Prolonged Strain, indicating that elevated load frequently occurred as part of sustained periods rather than only isolated spikes.
5. The 7-day and 14-day moving averages provided complementary perspectives on system behaviour. The 7-day average captured relatively recent changes, whereas the 14-day average provided a smoother representation of longer-duration load patterns.
6. The Tableau dashboard reported an AGG Backlog Accumulation Rate of 33.06%. Because this indicator is based on changes between consecutive System Load observations and is displayed as an aggregated table calculation, it should be interpreted as an analytical dashboard measure rather than as a direct 33.06% increase in total backlog over the entire study period.

Overall, the findings show that analysing System Load alone is insufficient to understand operational patterns fully. Combining Net Daily Intake, High Load classification, duration-based strain measures, moving averages, and Backlog Accumulation Rate provides a more comprehensive view of short-term fluctuations and sustained operational pressure.

8. RECOMMENDATIONS

Based on the analytical findings, the following recommendations are proposed to improve operational monitoring and support data-driven management of child custody system load.

1. **Continuous System Load Monitoring:** System Load should be monitored regularly using both daily values and moving averages. The 7-day moving average can support short-term monitoring, while the 14-day moving average can help identify more persistent changes in operational pressure.
2. **Early Warning for Sustained High Load:** Since the analysis identified 429 high-load observations and 411 prolonged-strain observations, duration-based alerts can be incorporated into monitoring systems. Consecutive above-average load observations can serve as an early indication of sustained operational pressure.
3. **Separate Monitoring of CBP and HHS Loads:** CBP custody and HHS care should continue to be analysed separately alongside total System Load. Their substantially different average levels demonstrate that an aggregate indicator alone may not reveal how operational load is distributed between the two components.
4. **Integrate Net Intake with System Load:** Net Daily Intake should be monitored together with System Load rather than independently. Sustained positive net intake combined with elevated System Load may provide an early indication of increasing operational pressure.
5. **Use Backlog Indicators with Trend Analysis:** Backlog Accumulation Rate should be interpreted together with its time-series trend and other operational measures. This reduces the risk of drawing conclusions from a single aggregated percentage.
6. **Interactive Dashboard-Based Monitoring:** The Tableau dashboard can be used as a monitoring framework by combining KPI cards, date filters, System Load trends, CBP-HHS comparisons, Net Daily Intake, moving averages, and strain indicators. This enables users to examine different periods and operational measures within a single analytical interface.

9. LIMITATIONS AND FUTURE SCOPE

9.1 Limitations

The study provides an analytical view of System Load, Net Daily Intake, backlog patterns, and sustained strain; however, several limitations should be considered while interpreting the results.

First, the analysis is based primarily on operational time-series data related to CBP custody and HHS care. The dataset does not contain detailed explanatory variables that could establish why particular increases or decreases in System Load occurred. Therefore, the study identifies patterns and associations rather than causal relationships.

Second, missing or unavailable observations within the study period may affect the continuity of time-series analysis. These observations were retained rather than automatically removed to avoid unnecessary loss or distortion of the original operational data.

Third, the High Load threshold was defined as System Load above the overall dataset mean. This provides a consistent analytical benchmark but does not represent an externally established operational capacity limit.

Fourth, the Backlog Accumulation Rate of 33.06% is an aggregated Tableau calculation derived from changes between consecutive System Load observations. Therefore, it should not be interpreted as a direct percentage increase in total backlog across the complete study period.

Finally, the study focuses on quantitative operational patterns and does not incorporate qualitative factors or external events that may have influenced custody and care levels.

9.2 Future Scope

Future research can extend the analysis by incorporating additional variables such as operational capacity, processing or transfer times, regional information, and other relevant contextual factors where reliable data are available. Such variables could provide a deeper understanding of the factors associated with changes in System Load.

Advanced time-series techniques could also be explored to identify seasonality, structural changes, and recurring patterns. Forecasting models may be developed and validated to estimate future System Load and support capacity planning.

Future studies could additionally develop more sophisticated early-warning indicators by combining Net Daily Intake, moving averages, consecutive High Load duration, and backlog measures. This could provide a more comprehensive framework for detecting emerging periods of operational strain.

10. CONCLUSION

This study applied data analytics and visualization techniques to examine System Load, Net Daily Intake, backlog patterns, and sustained operational strain within U.S. child custody operations involving CBP custody and HHS care during the study period from 12 January 2023 to 21 December 2025.

The analysis found an average System Load of approximately 6,233 children, with the maximum reaching 11,762 on 20 December 2023 and the minimum falling to 2,002 on 24 August 2025. Component-level analysis showed an average of 171.5 children in CBP custody and 6,061 children in HHS care, indicating that HHS care represented the substantially larger component of the measured System Load.

Net Daily Intake showed considerable short-term variation, ranging from +206 to -465, with an overall average of -44.74 children. The High Load analysis identified 429 above-average-load observations, while 411 observations were classified as Prolonged Strain. The use of 7-day and 14-day moving averages further supported the identification of short-term fluctuations and more persistent load patterns.

The study also incorporated a Backlog Accumulation Rate to examine relative changes in System Load between consecutive observations. Tableau reported an aggregated value of 33.06%, which was treated as an analytical dashboard indicator rather than a direct measure of total backlog growth across the complete study period.

Overall, the study demonstrates how Python, Pandas, and Tableau can be integrated to transform operational data into structured indicators and interactive visualizations. The resulting analytical framework enables System Load, intake patterns, and sustained strain to be examined together, providing a structured approach to operational monitoring and data-driven analysis.